

## Steps of the Machine Learning

- Data Collection
- EDA
  - Data Cleaning:
  - Data Analysis
  - Pre-processing
- Split the Data in to Training and Testing
- Transform
  - Encoding
  - Scaling
- Model Building
  - Model training
    - Model Evaluation
  - Prediction Pipeline

# Model Building

Training

Prediction

## Training Phase

- a) Choose model based on problem
- b) Hyper parameter tuning
- c) Model evaluation
- d) Select the best model
- e) Save model as pkl file

## Prediction Phase

- a) Create the API to predict the new data
- b) Handover the API to user (aws, or any other cloud)
- c) API ready for prediction

| x |  |  |  |  |  | y |
|---|--|--|--|--|--|---|
|---|--|--|--|--|--|---|

| CustID | Age | Monthly Spend | Tenure Months | Support Calls | Payment Method   | Return |
|--------|-----|---------------|---------------|---------------|------------------|--------|
| 1001   | 56  | 79.7          | 22            | 3             | Bank Transfer    | Yes    |
| 1002   | 46  | 63.38         | 44            | 0             | Bank Transfer    | No     |
| 1003   | 32  | 38.57         | 25            | 2             | Bank Transfer    | No     |
| 1004   | 60  | 104.62        | 27            | 4             | Credit Card      | No     |
| 1005   | 25  | 27.33         | 42            | 2             | Bank Transfer    | Yes    |
| 1006   | 38  | 113.86        | 28            | 4             | Credit Card      | No     |
| 1007   | 56  | 142.01        | 16            | 0             | Bank Transfer    | Yes    |
| 1008   | 36  | 20.1          | 15            | 1             | Electronic Check | No     |
| 1009   | 40  | 148.99        | 47            | 3             | Electronic Check | No     |
| 1010   | 28  | 100.27        | 44            | 0             | Credit Card      | Yes    |

## - EDA

- 1) Data Cleaning:
- 2) Data Analysis
- 3) Pre-processing

X= Independent Variable

Y= Dependent Variable

| x      |     |               |               |               |                  | y      |
|--------|-----|---------------|---------------|---------------|------------------|--------|
| CustID | Age | Monthly Spend | Tenure Months | Support Calls | Payment Method   | Return |
| 1001   | 56  | 79.7          | 22            | 3             | Bank Transfer    | Yes    |
| 1002   | 46  | 63.38         | 44            | 0             | Bank Transfer    | No     |
| 1003   | 32  | 38.57         | 25            | 2             | Bank Transfer    | No     |
| 1004   | 60  | 104.62        | 27            | 4             | Credit Card      | No     |
| 1005   | 25  | 27.33         | 42            | 2             | Bank Transfer    | Yes    |
| 1006   | 38  | 113.86        | 28            | 4             | Credit Card      | No     |
| 1007   | 56  | 142.01        | 16            | 0             | Bank Transfer    | Yes    |
| 1008   | 36  | 20.1          | 15            | 1             | Electronic Check | No     |
| 1009   | 40  | 148.99        | 47            | 3             | Electronic Check | No     |
| 1010   | 28  | 100.27        | 44            | 0             | Credit Card      | Yes    |

X= Independent Variable  
Y= Dependent Variable

### X\_train

| CustID | Age | Monthly Spend | Tenure Months | Support Calls | Payment Method |
|--------|-----|---------------|---------------|---------------|----------------|
| 1001   | 56  | 79.7          | 22            | 3             | Bank Transfer  |
| 1002   | 46  | 63.38         | 44            | 0             | Bank Transfer  |
| 1003   | 32  | 38.57         | 25            | 2             | Bank Transfer  |
| 1004   | 60  | 104.62        | 27            | 4             | Credit Card    |
| 1005   | 25  | 27.33         | 42            | 2             | Bank Transfer  |
| 1006   | 38  | 113.86        | 28            | 4             | Credit Card    |
| 1007   | 56  | 142.01        | 16            | 0             | Bank Transfer  |

### y\_train\_true

| Return |
|--------|
| Yes    |
| No     |
| No     |
| No     |
| Yes    |
| No     |
| Yes    |

- Split the Data in to Training and Testing
  - Training 70%
  - Testing 30%

### X\_test

| CustID | Age | Monthly Spend | Tenure Months | Support Calls | Payment Method   |
|--------|-----|---------------|---------------|---------------|------------------|
| 1008   | 36  | 20.1          | 15            | 1             | Electronic Check |
| 1009   | 40  | 148.99        | 47            | 3             | Electronic Check |
| 1010   | 28  | 100.27        | 44            | 0             | Credit Card      |

### y\_test\_true

| Retrun |
|--------|
| No     |
| No     |
| Yes    |

Data After Encoding

x\_train

| Age | Monthly Spend | Tenure Months | Support Calls | Bank Transfer | Credit Card | Electronic Check |
|-----|---------------|---------------|---------------|---------------|-------------|------------------|
| 56  | 79.7          | 22            | 3             | 1             | 0           | 0                |
| 46  | 63.38         | 44            | 0             | 1             | 0           | 0                |
| 32  | 38.57         | 25            | 2             | 1             | 0           | 0                |
| 60  | 104.62        | 27            | 4             | 0             | 1           | 0                |
| 25  | 27.33         | 42            | 2             | 1             | 0           | 0                |
| 38  | 113.86        | 28            | 4             | 0             | 1           | 0                |
| 56  | 142.01        | 16            | 0             | 1             | 0           | 0                |

| x_test |        |    |   |   |   |   |
|--------|--------|----|---|---|---|---|
| 36     | 20.1   | 15 | 1 | 0 | 0 | 1 |
| 40     | 148.99 | 47 | 3 | 0 | 0 | 1 |
| 28     | 100.27 | 44 | 0 | 0 | 1 | 0 |

## Scaling

| X_Train |               |               |               |               |             |                  |
|---------|---------------|---------------|---------------|---------------|-------------|------------------|
| Age     | Monthly Spend | Tenure Months | Support Calls | Bank Transfer | Credit Card | Electronic Check |
| 0.91    | -0.04         | -0.75         | 0.55          | 1             | 0           | 0                |
| 0.1     | -0.47         | 1.56          | -1.38         | 1             | 0           | 0                |
| -1.02   | -1.11         | -0.44         | -0.09         | 1             | 0           | 0                |
| 1.23    | 0.61          | -0.23         | 1.2           | 0             | 1           | 0                |
| -1.58   | -1.4          | 1.35          | -0.09         | 1             | 0           | 0                |
| -0.54   | 0.85          | -0.12         | 1.2           | 0             | 1           | 0                |
| 0.91    | 1.58          | -1.38         | -1.38         | 1             | 0           | 0                |

| X_test |               |               |               |               |             |                  |
|--------|---------------|---------------|---------------|---------------|-------------|------------------|
| Age    | Monthly Spend | Tenure Months | Support Calls | Bank Transfer | Credit Card | Electronic Check |
| 0.27   | -1.31         | -1.41         | -0.27         | 0             | 0           | 1                |
| 1.07   | 1.11          | 0.81          | 1.34          | 0             | 0           | 1                |
| -1.34  | 0.2           | 0.6           | -1.07         | 0             | 1           | 0                |

## x\_train

| Age   | Monthly Spend | Tenure Months | Support Calls | Bank Transfer | Credit Card | Electronic Check |
|-------|---------------|---------------|---------------|---------------|-------------|------------------|
| 0.91  | -0.04         | -0.75         | 0.55          | 1             | 0           | 0                |
| 0.1   | -0.47         | 1.56          | -1.38         | 1             | 0           | 0                |
| -1.02 | -1.11         | -0.44         | -0.09         | 1             | 0           | 0                |
| 1.23  | 0.61          | -0.23         | 1.2           | 0             | 1           | 0                |
| -1.58 | -1.4          | 1.35          | -0.09         | 1             | 0           | 0                |
| -0.54 | 0.85          | -0.12         | 1.2           | 0             | 1           | 0                |
| 0.91  | 1.58          | -1.38         | -1.38         | 1             | 0           | 0                |

Model  
Training

Model.pkl

**x\_train**

| Age   | Monthly Spend | Tenure Months | Support Calls | Bank Transfer | Credit Card | Electronic Check |
|-------|---------------|---------------|---------------|---------------|-------------|------------------|
| 0.91  | -0.04         | -0.75         | 0.55          | 1             | 0           | 0                |
| 0.1   | -0.47         | 1.56          | -1.38         | 1             | 0           | 0                |
| -1.02 | -1.11         | -0.44         | -0.09         | 1             | 0           | 0                |
| 1.23  | 0.61          | -0.23         | 1.2           | 0             | 1           | 0                |
| -1.58 | -1.4          | 1.35          | -0.09         | 1             | 0           | 0                |
| -0.54 | 0.85          | -0.12         | 1.2           | 0             | 1           | 0                |
| 0.91  | 1.58          | -1.38         | -1.38         | 1             | 0           | 0                |



Model.pkl



**y\_train\_pred**

| Return |
|--------|
| 1      |
| 1      |
| 1      |
| 0      |
| 1      |
| 0      |
| 1      |

training\_x

| Age   | Monthly Spend | Tenure Months | Support Calls | Bank Transfer | Credit Card | Electronic Check |
|-------|---------------|---------------|---------------|---------------|-------------|------------------|
| 0.91  | -0.04         | -0.75         | 0.55          | 1             | 0           | 0                |
| 0.1   | -0.47         | 1.56          | -1.38         | 1             | 0           | 0                |
| -1.02 | -1.11         | -0.44         | -0.09         | 1             | 0           | 0                |
| 1.23  | 0.61          | -0.23         | 1.2           | 0             | 1           | 0                |
| -1.58 | -1.4          | 1.35          | -0.09         | 1             | 0           | 0                |
| -0.54 | 0.85          | -0.12         | 1.2           | 0             | 1           | 0                |
| 0.91  | 1.58          | -1.38         | -1.38         | 1             | 0           | 0                |

y train\_true

| Return |
|--------|
| 1      |
| 0      |
| 0      |
| 0      |
| 1      |
| 0      |
| 1      |

y\_train\_pred

| Return |
|--------|
| 1      |
| 1      |
| 1      |
| 0      |
| 1      |
| 0      |
| 1      |

**x\_test**

| Age   | Monthly Spend | Tenure Months | Support Calls | Bank Transfer | Credit Card | Electronic Check |
|-------|---------------|---------------|---------------|---------------|-------------|------------------|
| 0.27  | -1.31         | -1.41         | -0.27         | 0             | 0           | 1                |
| 1.07  | 1.11          | 0.81          | 1.34          | 0             | 0           | 1                |
| -1.34 | 0.2           | 0.6           | -1.07         | 0             | 1           | 0                |

**y\_test\_true**

| Return |
|--------|
| 0      |
| 0      |
| 1      |



Model.pkl



**y\_test\_pred**

| Return |
|--------|
| 1      |
| 0      |
| 1      |

**x\_train**

| Age   | Monthly Spend | Tenure Months | Support Calls | Bank Transfer | Credit Card | Electronic Check |
|-------|---------------|---------------|---------------|---------------|-------------|------------------|
| 0.91  | -0.04         | -0.75         | 0.55          | 1             | 0           | 0                |
| 0.1   | -0.47         | 1.56          | -1.38         | 1             | 0           | 0                |
| -1.02 | -1.11         | -0.44         | -0.09         | 1             | 0           | 0                |
| 1.23  | 0.61          | -0.23         | 1.2           | 0             | 1           | 0                |
| -1.58 | -1.4          | 1.35          | -0.09         | 1             | 0           | 0                |
| -0.54 | 0.85          | -0.12         | 1.2           | 0             | 1           | 0                |
| 0.91  | 1.58          | -1.38         | -1.38         | 1             | 0           | 0                |

**Y\_train\_true**

| Return |
|--------|
| 1      |
| 0      |
| 0      |
| 0      |
| 1      |
| 0      |
| 1      |

**Y\_train\_pred**

| Return |
|--------|
| 1      |
| 1      |
| 1      |
| 0      |
| 1      |
| 0      |
| 1      |

**x\_test**

| Age   | Monthly Spend | Tenure Months | Support Calls | Bank Transfer | Credit Card | Electronic Check |
|-------|---------------|---------------|---------------|---------------|-------------|------------------|
| 0.27  | -1.31         | -1.41         | -0.27         | 0             | 0           | 1                |
| 1.07  | 1.11          | 0.81          | 1.34          | 0             | 0           | 1                |
| -1.34 | 0.2           | 0.6           | -1.07         | 0             | 1           | 0                |

**y\_test\_true**

| Return |
|--------|
| 0      |
| 0      |
| 1      |

**Y\_test\_pred**

| Return |
|--------|
| 1      |
| 0      |
| 1      |

Evaluation Phase

Training Data Evaluation

Testing Data Evaluation

Training  
Data Evaluation

Actual\_  
training\_y

| Return |
|--------|
| 1      |
| 0      |
| 0      |
| 0      |
| 1      |
| 0      |
| 1      |

Training\_  
predicted\_y

| Return |
|--------|
| 1      |
| 1      |
| 1      |
| 0      |
| 1      |
| 0      |
| 1      |

Testing  
Data Evaluation

Actual\_  
testing\_y

| Return |
|--------|
| 0      |
| 0      |
| 1      |

predicted\_  
testing\_y

| Return |
|--------|
| 1      |
| 0      |
| 1      |

## Data Collection

### EDA

- 1) Data Cleaning:
  - 2) Data Analysis
  - 3) Advance Pre-processing
    - Handel Imbalance Data
    - Feature Selection
    - Dimension Reduction
  - 4) Encoding
  - 5) Split the data Training and Testing
    - Training Data 80%,
    - Testing Data 20%
- Scaling
  - Model Building
    - Model training
    - Model Evaluation
    - Prediction Pipeline